Keast Group Protocols

Adult Mouse DRG Dissociation for FACS of TrpV1$^{PLAP\ nLacZ}$ Mice

Set up

1. Turn on hood: UV for 15 mins, then laminar flow for 5 mins.
2. Put all the reagents required on ice to thaw: collagenase, trypsin/EDTA.
3. Put dissection instruments in 80% ethanol 10 mins then allow to dry
4. Put 2.5ml 1X Tyrodes in 2 x 35mm petri dishes per mouse and put on ice.
5. Put 3.5ml 1X Tyrodes in 2 x 15ml falcon tubes per mouse and put on ice.
6. Add 320µl 1X Tyrodes to 750µl collagenase (per sample). Pre-warm the solution to 37°C.

Dissection

1. Following the approved method for euthanizing the animal, dissect ganglia and place each in 1X Tyrodes in a 35mm petri dish on ice until the end of the dissection.

Dissociation

1. Using forceps, transfer ganglia into a 15 ml tube containing 3.5 ml 1X Tyrodes. (**Perform remaining steps in the laminar flow**)
2. Allow ganglia to settle and remove the 1X Tyrodes solution.
3. Add pre-warmed collagenase/Tyrodes to the tube. Give the tube a gentle flick to mix and place in an incubator at 37°C for 1 hour.
4. While this is incubating, pre-warm trypsin/EDTA solution to 37°C (320µl trypsin/EDTA + 750µl collagenase [per sample])
5. Allow tissue to settle and remove collagenase/Tyrodes solution. Add collagenase/trypsin/EDTA solution and mix gently. Incubate at 37°C for 1 hour.
6. Allow tissue to settle, remove supernatant and discard.
7. Wash twice with 1ml Leibovitz's L-15 medium (Life Technologies, Cat# 21083-027): i.e., with transfer pipette, add 1 ml L-15, allow tissue to settle and discard supernatant. Repeat.
8. Triturate cells in 1ml L-15 medium. Do this by triturating 20 times with a 1000µl pipette.
9. Centrifuge at 300x g for 8 mins at 4°C.
10. Remove and discard supernatant using a pipette being careful not to disturb the cell pellet.
11. Resuspend cells in 500 µl L-15 medium.

12. Add 5μl of 1mM DDAOG (Molecular Probes, Cat# D-6488) in DMSO to the dissociated cells, mix gently and incubate on ice for 30 mins.
13. Filter the dissociated cells through a 40μm pore size cell strainer and collect the cells into a FACS tube.
14. Add 1μl of 100 μg/mL DAPI (200ng/ml final concentration) as a dead cell stain just prior to taking the samples for FACS.

FACS Sorting and RNA Extraction

1. Take cells to the FACS facility for sorting. Keep the cells on ice.
2. Sort cells directly into 1.5ml eppendorf tubes containing 500μl Qiagen RLT lysis buffer + 30μl 2M DTT. Keep tubes on ice after sorting.
3. Homogenize cells by passing them through a 25 gauge needle 3 times. Proceed to RNA extraction immediately.
4. Extract RNA using the Qiagen RNeasy Plus Micro kit according to the manufacturer’s instructions with the exception of adding 400 μl 70% ethanol per sample at Step 5 (long protocol)/ Step 3 (quick protocol).
5. Store the RNA at -80°C.

Reagents

Tyrodes solution
10x stock

1. Mix the following:
   7.59 g NaCl
   1.68 g $NaHCO_3$
   2.38 g HEPES
   224 mg KCl (or 3 ml 1 M KCl)
2. Make up to 100 ml with H_2O and autoclave. Store at 4°C.

1x Tyrodes

1. Mix the following:
   500 µl antibiotic/antimycotic solution (0.5%)
   1.2 ml 1 M glucose (filter sterilized)
   400 µl 1 M $CaCl_2$ (filter sterilized)
   100 µl 1 M $MgCl_2$ (filter sterilized)
2. Add 87.8 ml autoclaved H_2O and 10 ml of 10X stock.
3. Store remaining solution at 4°C for $\leq$ 1 month.

Antibiotic/antimycotic solution (Invitrogen, cat# 15240-062)

- Dispense into 500 µl aliquots, store at -20°C

Collagenase, type 1 (Supplier: Scima R, Manufacturer: Worthington, Cat# L5004196)

- 20 mg collagenase in 15 ml Tyrodes, filter sterilize, aliquot into 750 µl, store at -20°C.

Trypsin/EDTA (Gibco, Cat #25200056, 100 ml, 0.25% trypsin + 1 mM EDTA or Sigma T4049, 0.25% trypsin/EDTA)

- Dispense into 650 µl aliquots, store at -20°C.